Algebra and Discrete Mathematics (ADM)

Tutorial 10 Paths and spanning trees

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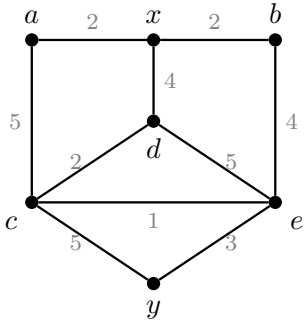
Dijkstra's Algorithm – Q1.3(a)

Question

Apply Dijkstra's Algorithm to Question 1.3(a), with start x and end y.



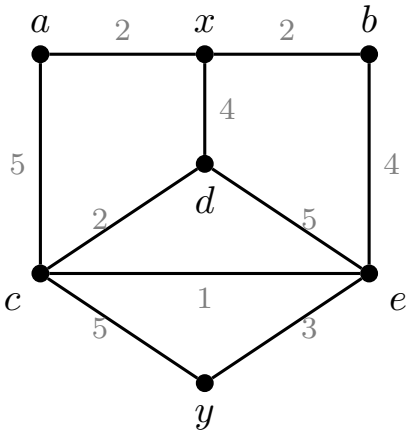
Dijkstra's Algorithm – Q1.3(a)

- Step 1: highlight x
- Labels:

$$L(v) = \begin{cases} (-, 0), & v = x \\ (-, \infty), & \text{otherwise} \end{cases}$$

- $F = \{a, b, c, d, e, y\}$
- Current vertex: x

x	a	b	c	d	e	y
0	∞	∞	∞	∞	∞	∞



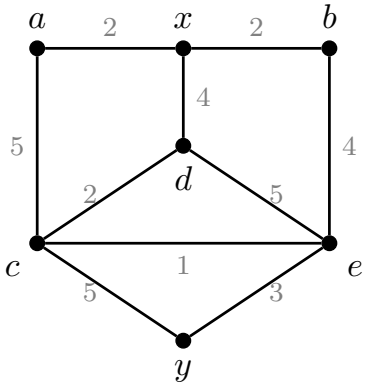
Dijkstra's Algorithm – Q1.3(a)

- Step 2: from x, update neighbors a, b, d

$$L(a) = (x, 2), \quad L(b) = (x, 2), \quad L(d) = (x, 4)$$

- Step 3: smallest weight is 2, at a or b. Choose b and highlight xb. New current vertex: b

	a	b	c	d	e	y
	∞	∞	∞	∞	∞	∞
$x(0)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	∞	∞



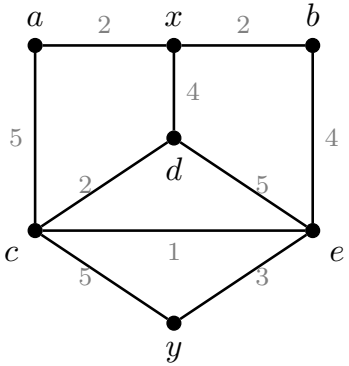
Dijkstra's Algorithm – Q1.3(a)

- Step 2: from b, update e

$$\omega(b) + \omega(be) = 2 + 4 = 6 \Rightarrow L(e) = (b, 6)$$

- Step 3: smallest weight is at a. Highlight a and edge xa. New current vertex: a

	a	b	c	d	e	y
	∞	∞	∞	∞	∞	∞
$x(0)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	∞	∞
$b(2)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	$(b, 6)$	∞



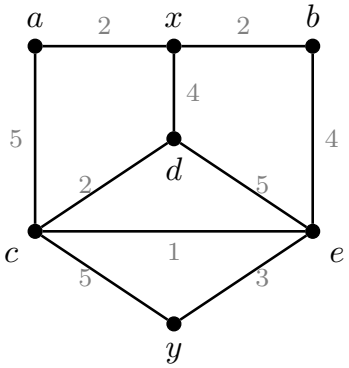
Dijkstra's Algorithm – Q1.3(a)

- Step 2: from a, update c

$$\omega(a) + \omega(ac) = 2 + 5 = 7 \Rightarrow L(c) = (a, 7)$$

- Step 3: smallest weight is at d. Highlight d and edge xd. New current vertex: d

	a	b	c	d	e	y
	∞	∞	∞	∞	∞	∞
$x(0)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	∞	∞
$b(2)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	$(b, 6)$	∞
$a(2)$	$(x, 2)$	$(x, 2)$	$(a, 7)$	$(x, 4)$	$(b, 6)$	∞



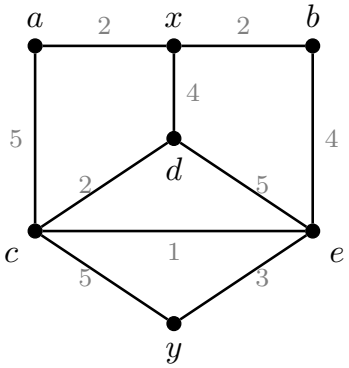
Dijkstra's Algorithm – Q1.3(a)

- Step 2: from d, improve the label of c

$$\omega(d) + \omega(dc) = 4 + 2 = 6 < 7 \Rightarrow L(c) = (d, 6)$$

- Step 3: smallest temporary weight is 6, at c or e. Choose e, highlight be. New current vertex: e

	a	b	c	d	e	y
	∞	∞	∞	∞	∞	∞
$x(0)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	∞	∞
$b(2)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	$(b, 6)$	∞
$a(2)$	$(x, 2)$	$(x, 2)$	$(a, 7)$	$(x, 4)$	$(b, 6)$	∞
$d(4)$	$(x, 2)$	$(x, 2)$	$(d, 6)$	$(x, 4)$	$(b, 6)$	∞



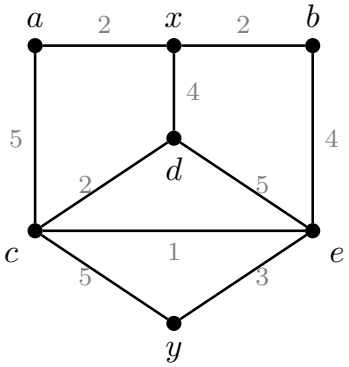
Dijkstra's Algorithm – Q1.3(a)

- Step 2: from e, update y

$$\omega(e) + \omega(ey) = 6 + 3 = 9 \Rightarrow L(y) = (e, 9)$$

- Step 3: the smallest temporary weight is at c. Highlight c and edge dc. New current vertex: c

	a	b	c	d	e	y
	∞	∞	∞	∞	∞	∞
$x(0)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	∞	∞
$b(2)$	$(x, 2)$	$(x, 2)$	∞	$(x, 4)$	$(b, 6)$	∞
$a(2)$	$(x, 2)$	$(x, 2)$	$(a, 7)$	$(x, 4)$	$(b, 6)$	∞
$d(4)$	$(x, 2)$	$(x, 2)$	$(d, 6)$	$(x, 4)$	$(b, 6)$	∞
$e(6)$	$(x, 2)$	$(x, 2)$	$(d, 6)$	$(x, 4)$	$(b, 6)$	$(e, 9)$



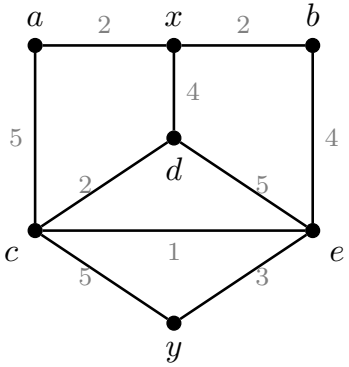
Dijkstra's Algorithm – Q1.3(a)

- Step 2: from c,

$$\omega(c) + \omega(cy) = 6 + 5 = 11 > 9$$

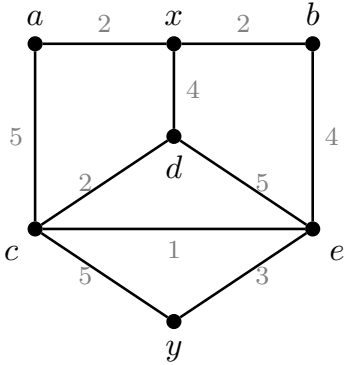
  so the label of y does not change
- Step 3: highlight y and stop

	a	b	c	d	e	y
	∞	∞	∞	∞	∞	∞
$x(0)$	$(x,2)$	$(x,2)$	∞	$(x,4)$	∞	∞
$b(2)$	$(x,2)$	$(x,2)$	∞	$(x,4)$	$(b,6)$	∞
$a(2)$	$(x,2)$	$(x,2)$	$(a,7)$	$(x,4)$	$(b,6)$	∞
$d(4)$	$(x,2)$	$(x,2)$	$(d,6)$	$(x,4)$	$(b,6)$	∞
$e(6)$	$(x,2)$	$(x,2)$	$(d,6)$	$(x,4)$	$(b,6)$	$(e,9)$
$c(6)$	$(x,2)$	$(x,2)$	$(d,6)$	$(x,4)$	$(b,6)$	$(e,9)$



Dijkstra's Algorithm – Q1.3(a)

	a	b	c	d	e	y
	∞	∞	∞	∞	∞	∞
$x(0)$	$(x,2)$	$(x,2)$	∞	$(x,4)$	∞	∞
$b(2)$	$(x,2)$	$(x,2)$	∞	$(x,4)$	$(b,6)$	∞
$a(2)$	$(x,2)$	$(x,2)$	$(a,7)$	$(x,4)$	$(b,6)$	∞
$d(4)$	$(x,2)$	$(x,2)$	$(d,6)$	$(x,4)$	$(b,6)$	∞
$e(6)$	$(x,2)$	$(x,2)$	$(d,6)$	$(x,4)$	$(b,6)$	$(e,9)$
$c(6)$	$(x,2)$	$(x,2)$	$(d,6)$	$(x,4)$	$(b,6)$	$(e,9)$



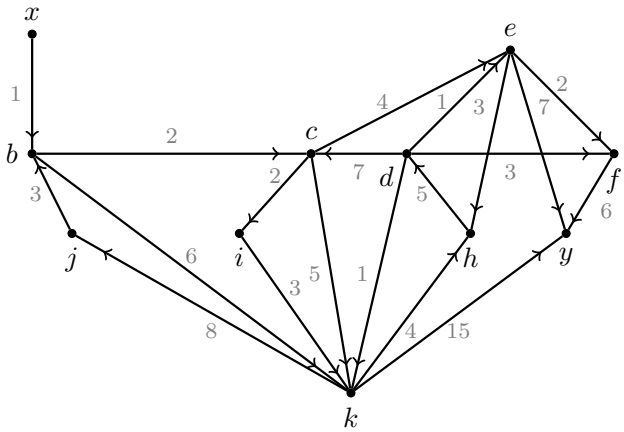
$$y \leftarrow e \leftarrow b \leftarrow x$$

$$\boxed{x \to b \to e \to y, \quad \text{weight } 9}$$

Dijkstra's Algorithm for digraphs – Q3.3(a)

Question

Apply Dijkstra's Algorithm to Question 3.3(a), with start x and end y.



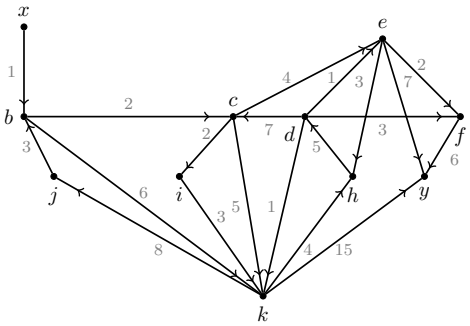
Dijkstra's Algorithm for digraphs – Q3.3(a)

- Step 1: highlight x
- Labels:

$$L(v) = \begin{cases} (-, 0), & v = x \\ (-, \infty), & \text{otherwise} \end{cases}$$

- $F = \{a, b, c, d, e, f, g, h, i, j, k, y\}$
- Current vertex: x

x	a	b	c	d	e	f	g	h	i	j	k	y
0	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞

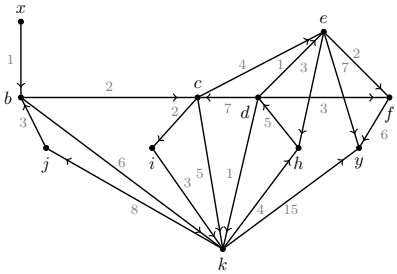


Dijkstra's Algorithm for digraphs – Q3.3(a)

- Step 2: from x, the only out-neighbor is b

$$\omega(x) + \omega(xb) = 0 + 1 = 1 \Rightarrow L(b) = (x, 1)$$

- Step 3: highlight b and edge xb. New current vertex: b



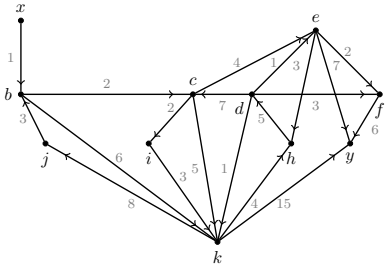
	a	b	c	d	e	f	g	h	i	j	k	y
	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$x(0)$	∞	$(x, 1)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞

Dijkstra's Algorithm for digraphs – Q3.3(a)

- Step 2: from b, update out-neighbors c, j, k

$$L(c) = (b, 3), \quad L(j) = (b, 4), \quad L(k) = (b, 9)$$

- Step 3: highlight c and edge bc. New current vertex: c



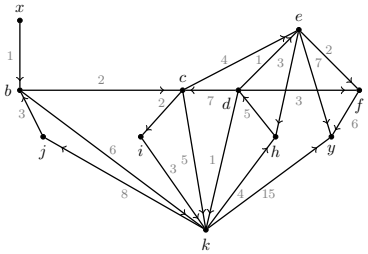
	a	b	c	d	e	f	g	h	i	j	k	y
	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$x(0)$	∞	$(x, 1)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$b(1)$	∞	$(x, 1)$	$(b, 3)$	∞	∞	∞	∞	∞	∞	$(b, 4)$	$(b, 9)$	∞

Dijkstra's Algorithm for digraphs – Q3.3(a)

- Step 2: from c, update e, i, k

$$L(e) = (c, 7), \quad L(i) = (c, 5), \quad L(k) = (c, 8)$$

- Step 3: highlight j
- Then Step 2 from j: no label improves
- Step 3: highlight i



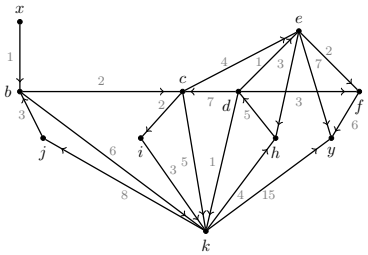
	a	b	c	d	e	f	g	h	i	j	k	y
	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$x(0)$	∞	$(x,1)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$b(1)$	∞	$(x,1)$	$(b,3)$	∞	∞	∞	∞	∞	∞	$(b,4)$	$(b,9)$	∞
$c(3)$	∞	$(x,1)$	$(b,3)$	∞	$(c,7)$	∞	∞	∞	$(c,5)$	$(b,4)$	$(c,8)$	∞
$j(4)$	∞	$(x,1)$	$(b,3)$	∞	$(c,7)$	∞	∞	∞	$(c,5)$	$(b,4)$	$(c,8)$	∞

Dijkstra's Algorithm for digraphs – Q3.3(a)

- Step 2: from i, k stays at weight 8
- Step 3: highlight e
- Step 2: from e, update

$$L(f) = (e, 9), \quad L(h) = (e, 10), \quad L(y) = (e, 14)$$

- Step 3: highlight k
- Step 2: from k, update d: $L(d) = (k, 9)$



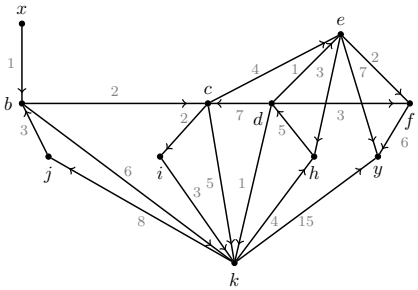
	a	b	c	d	e	f	g	h	i	j	k	y
	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$x(0)$	∞	$(x,1)$	∞	∞	∞	∞	∞	∞	∞	∞	∞	∞
$b(1)$	∞	$(x,1)$	$(b,3)$	∞	∞	∞	∞	∞	∞	$(b,4)$	$(b,9)$	∞
$c(3)$	∞	$(x,1)$	$(b,3)$	∞	$(c,7)$	∞	∞	∞	$(c,5)$	$(b,4)$	$(c,8)$	∞
$j(4)$	∞	$(x,1)$	$(b,3)$	∞	$(c,7)$	∞	∞	∞	$(c,5)$	$(b,4)$	$(c,8)$	∞
$i(5)$	∞	$(x,1)$	$(b,3)$	∞	$(c,7)$	∞	∞	∞	$(c,5)$	$(b,4)$	$(c,8)$	∞
$e(7)$	∞	$(x,1)$	$(b,3)$	∞	$(c,7)$	$(e,9)$	∞	$(e,10)$	$(c,5)$	$(b,4)$	$(c,8)$	$(e,14)$
$k(8)$	∞	$(x,1)$	$(b,3)$	$(k,9)$	$(c,7)$	$(e,9)$	∞	$(e,10)$	$(c,5)$	$(b,4)$	$(c,8)$	$(e,14)$

Dijkstra's Algorithm for digraphs – Q3.3(a)

- Continue alternating Step 3 and Step 2
- Highlight d, f, and h
- None of these improves the label of y
- Finally highlight y and stop

$$y \leftarrow e \leftarrow c \leftarrow b \leftarrow x$$

$$\boxed{x \to b \to c \to e \to y, \quad \text{weight } 14}$$

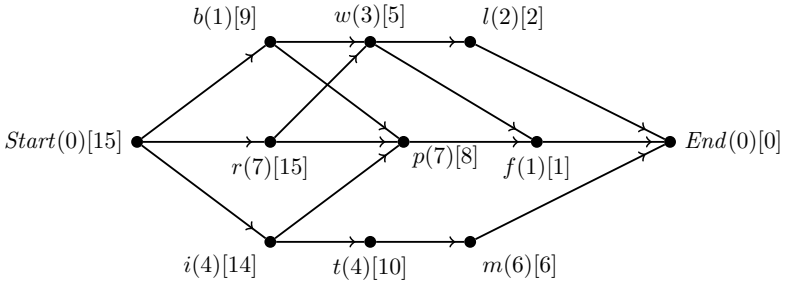


Critical path and project scheduling

- This spring, you are tackling the jungle that is your backyard. Some good friends have volunteered their time, and you have divided them into two groups.

Task	Vertex Name	Processing Time	Precedence Relationships
Buy plants	b	1	
Remove bushes	r	7	
Remove ivy	i	4	
Weed flower beds	w	3	b, r
Plant bushes	p	7	b, r
Plant flowers	f	1	w, p
Trim trees	t	4	i
Mow and rake lawn	m	6	t
Install lighting	l	2	w

Critical path and project scheduling



$\mathrm{ct}[l] = 2,$

$\mathrm{ct}[m] = 6,$

$\mathrm{ct}[t] = \mathrm{pt}(t) + \mathrm{ct}[m] = 4 + 6 = 10,$

$\mathrm{ct}[i] = \mathrm{pt}(i) + \mathrm{ct}[t] = 4 + 10 = 14,$

$\mathrm{ct}[b] = \mathrm{pt}(b) + \mathrm{ct}[p] = 1 + 8 = 9,$

$\mathrm{ct}[f] = 1,$

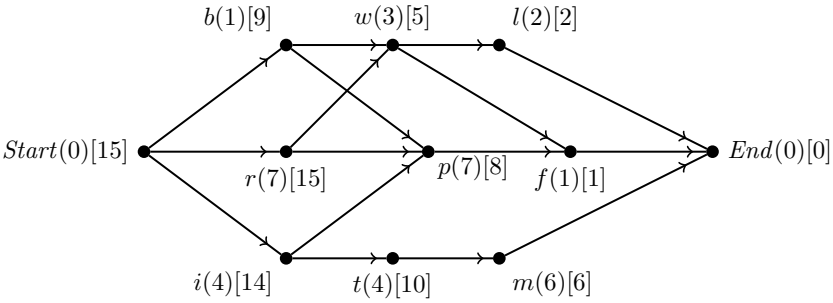
$\mathrm{ct}[p] = \mathrm{pt}(p) + \mathrm{ct}[f] = 7 + 1 = 8,$

$\mathrm{ct}[w] = \mathrm{pt}(w) + \mathrm{ct}[l] = 3 + 2 = 5,$

$\mathrm{ct}[r] = \mathrm{pt}(r) + \mathrm{ct}[p] = 7 + 8 = 15,$

$\mathrm{ct}[\mathit{Start}] = \mathrm{pt}(\mathit{Start}) + \mathrm{ct}[r] = 0 + 15 = 15$

Critical path and project scheduling



The critical path is

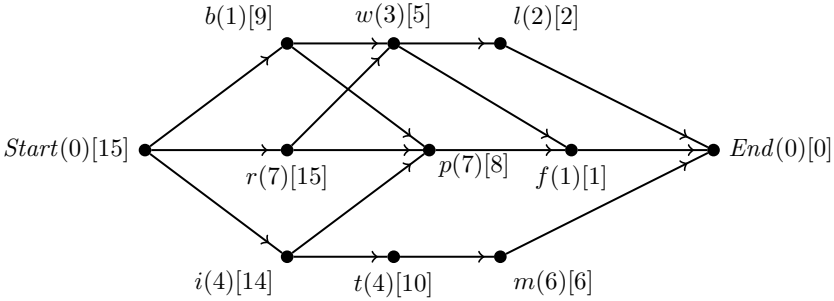
$$Start \rightarrow r \rightarrow p \rightarrow f \rightarrow End$$

The corresponding critical-path priority list is

$$r - i - t - b - p - m - w - l - f$$

Critical path and project scheduling

$r - i - t - b - p - m - w - l - f$

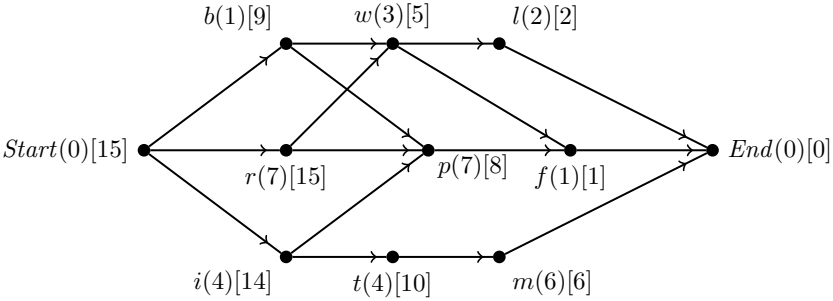


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
P_1	r	r	r	r	r	r	r												
P_2	i	i	i	i															

-
- Step 1. $(T = 0)$ Assign r to P_1 and i to P_2.

Critical path and project scheduling

$r - i - t - b - p - m - w - l - f$

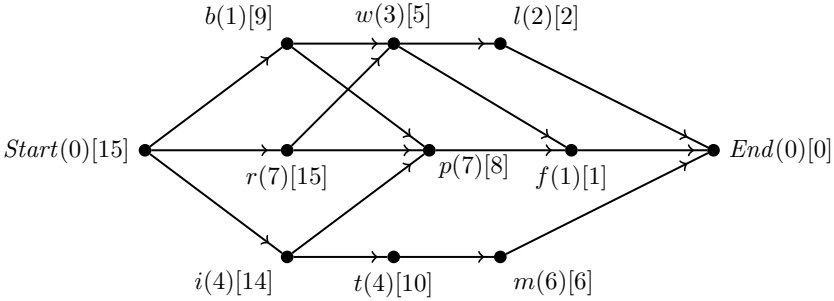


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
P_1	r	r	r	r	r	r	r												
P_2	i	i	i	i	t	t	t	t											

- Step 2. ($T = 4$) Assign t to P_2.

Critical path and project scheduling

$r - i - t - b - p - m - w - l - f$

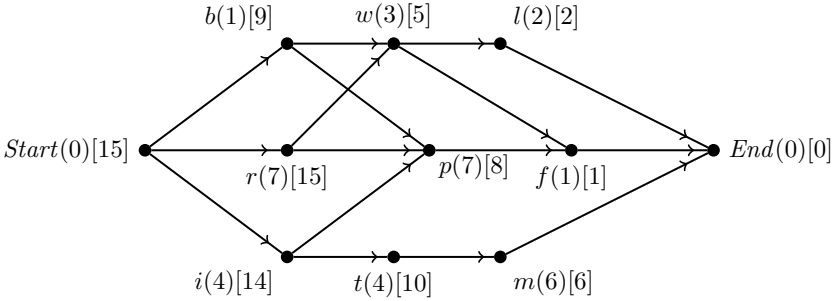


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
P_1	r	r	r	r	r	r	r	b											
P_2	i	i	i	i	t	t	t	t											

- Step 3. $(T = 7)$ Assign b to P_1.

Critical path and project scheduling

$r - i - t - b - p - m - w - l - f$

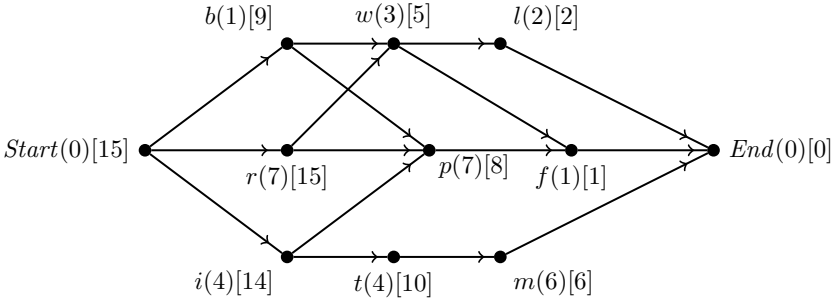


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
P_1	r	r	r	r	r	r	r	b	p	p	p	p	p	p	p				
P_2	i	i	i	i	t	t	t	t	m	m	m	m	m	m					

- Step 4. ($T = 8$) Assign p to P_1 and m to P_2.

Critical path and project scheduling

$r - i - t - b - p - m - w - l - f$

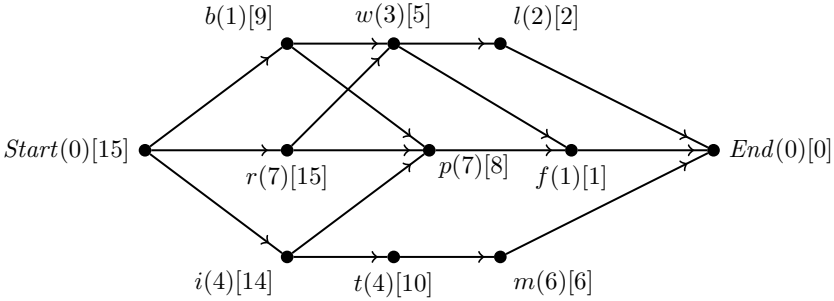


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
P_1	r	r	r	r	r	r	r	b	p	p	p	p	p	p	p				
P_2	i	i	i	i	t	t	t	t	m	m	m	m	m	m	w	w	w		

- Step 5. ($T = 14$) Assign w to P_2.

Critical path and project scheduling

$r - i - t - b - p - m - w - l - f$

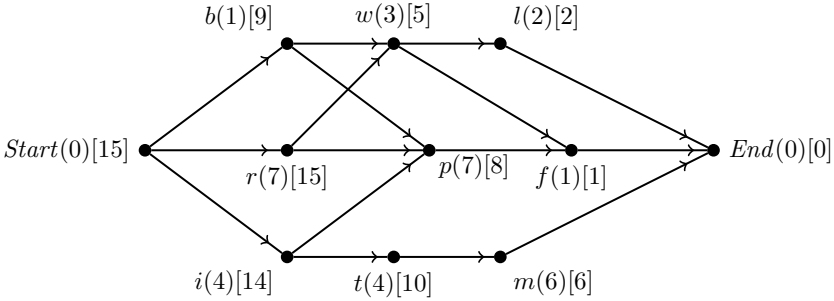


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
P_1	r	r	r	r	r	r	r	b	p	p	p	p	p	p	p	$*$	$*$		
P_2	i	i	i	i	t	t	t	t	m	m	m	m	m	m	w	w	w		

- Step 6. ($T = 15$) No task is eligible, so P_1 remains idle.

Critical path and project scheduling

$r - i - t - b - p - m - w - l - f$



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
P_1	r	r	r	r	r	r	r	b	p	p	p	p	p	p	p	$*$	$*$	l	l
P_2	i	i	i	i	t	t	t	t	m	m	m	m	m	m	w	w	w	f	$*$

- Step 7. ($T = 17$) Assign l to P_1 and f to P_2.

Critical path and project scheduling

- Using the two lower bounds described in the lecture, we obtain

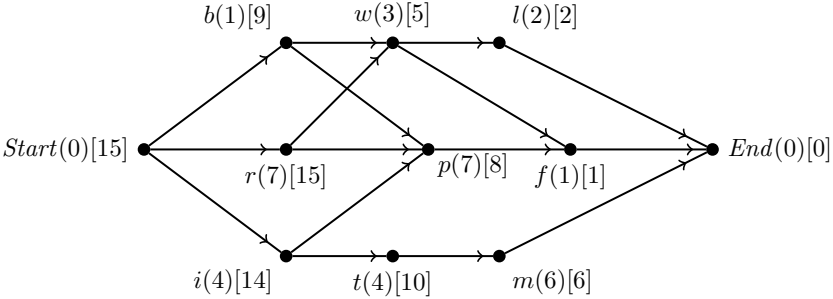
$$OPT \geq 15, \quad OPT \geq \frac{35}{2} = 17.5 \text{ for two processors}$$

- Since time is measured in whole units, the optimal finishing time must in fact be at least 18 hours.
- The schedule we have found contains 3 total hours of idle time.
- Task w does not appear high on the priority list, but it needed to be completed earlier because both l and f depend on it.
- An optimal schedule is as follows.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
P_1	r	r	r	r	r	r	r	p	p	p	p	p	p	p	l	l	f	$*$
P_2	i	i	i	i	b	t	t	t	t	w	w	w	m	m	m	m	m	m

What if we use 3 processors?

$r - i - t - b - p - m - w - l - f$

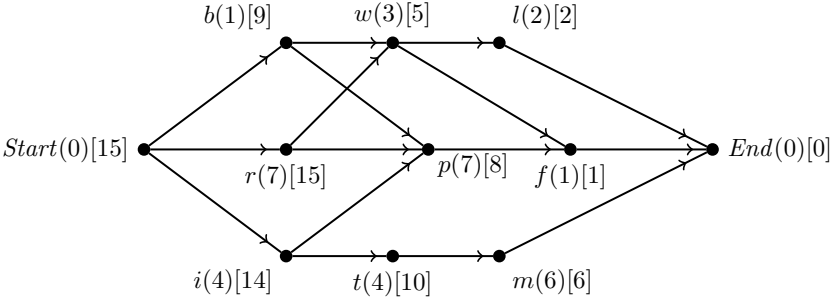


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
P_1	r	r	r	r	r	r	r											
P_2	i	i	i	i														
P_3	b	$*$	$*$	$*$														

At $T = 1$, no tasks are eligible.

What if we use 3 processors?

$r - i - t - b - p - m - w - l - f$

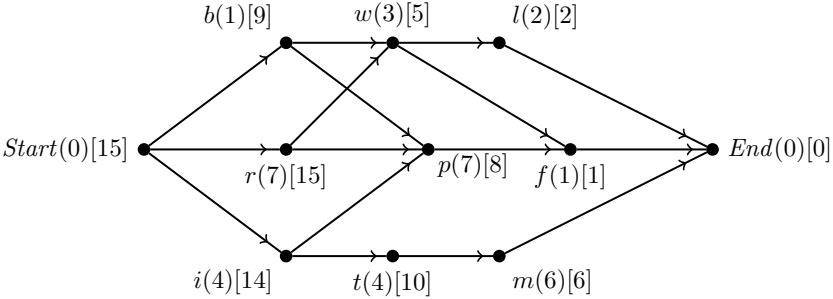


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
P_1	r	r	r	r	r	r	r	p	p	p	p	p	p	p				
P_2	i	i	i	i	t	t	t	t										
P_3	b	$*$	$*$	$*$	$*$	$*$	$*$	w	w	w								

At $T = 4$, the only eligible task is t. At $T = 7$, the eligible tasks are p and w.

What if we use 3 processors?

$r - i - t - b - p - m - w - l - f$

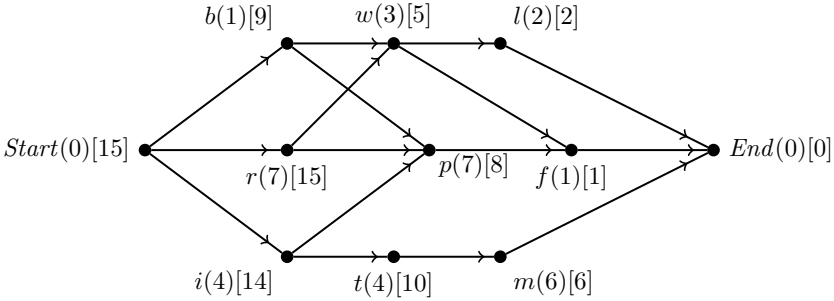


	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
P_1	r	r	r	r	r	r	r	p	p	p	p	p	p	p				
P_2	i	i	i	i	t	t	t	t	m	m	m	m	m	m				
P_3	b	$*$	$*$	$*$	$*$	$*$	$*$	w	w	w	l	l						

At $T = 8$, the next eligible task is m. At $T = 10$, the next eligible task is l.

What if we use 3 processors?

$r - i - t - b - p - m - w - l - f$



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
P_1	r	r	r	r	r	r	r	p	p	p	p	p	p	p	f			
P_2	i	i	i	i	t	t	t	t	m	m	m	m	m	m	$*$			
P_3	b	$*$	$*$	$*$	$*$	$*$	$*$	w	w	w	l	l	$*$	$*$	$*$			

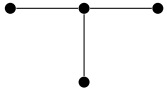
The final task f becomes eligible only after p has been completed. Thus, we obtain an optimal schedule with finishing time 15 hours.

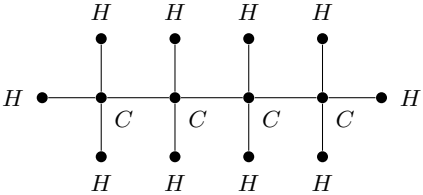
Tree – example

- Chemical Graph Theory uses concepts from graph theory to obtain results about chemical compounds
- Individual atoms in a molecule are represented by vertices, and an edge denotes a bond between two atoms.
- One way to determine the number of isomers (compounds with the same molecular formula but different arrangements of atoms) is to determine the number of distinct graphs with the correct types of vertices.
- For hydrocarbons, that is, molecules containing only carbon and hydrogen atoms, the hydrogen-depleted graph is often used, since the bonds between the carbon atoms uniquely determine the locations of the hydrogen atoms.

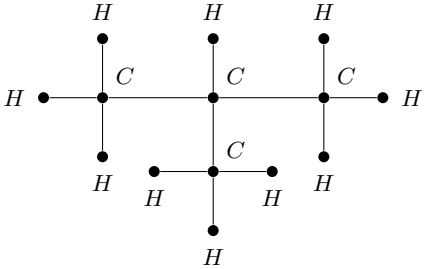
Tree – example

- There are only two trees on four vertices.
- Therefore, there are only two possible structural isomers of butane, C_4H_{10}.
- Graph theory shows that no other isomers of butane are possible, since no other trees on four vertices exist.





n-butane



isobutane

Kruskal's Algorithm – example

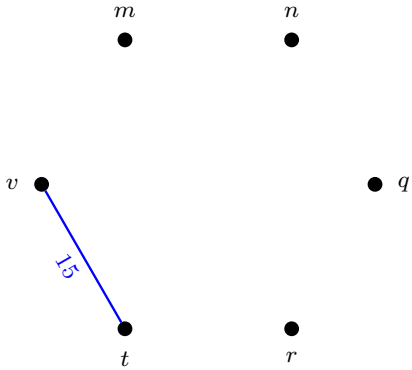
- The Optos Cable Company is expanding its fiber optic network over the next few years
- The company needs to lay new cable, but wishes to do so at minimal cost.
- Cost of the cable is fixed per meter
- The required cable lengths between pairs of towns are given in the table.

	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·

Kruskal's Algorithm – example

- Apply Kruskal's Algorithm; the edges can be selected directly from the table.
- The smallest weight is tv.

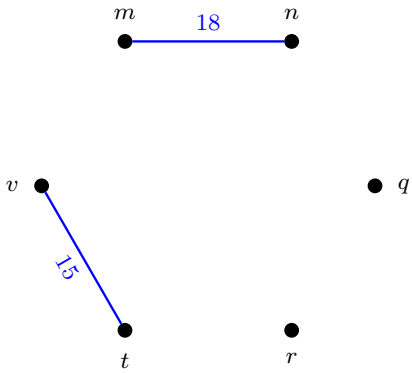
	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



Kruskal's Algorithm – example

- The next smallest weight is 18, corresponding to edge mn.

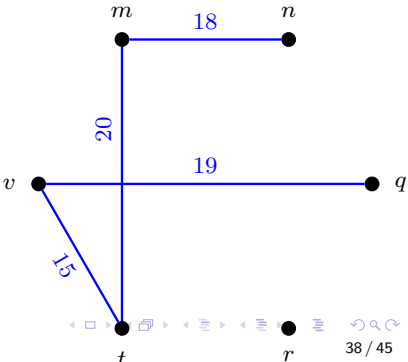
	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



Kruskal's Algorithm – example

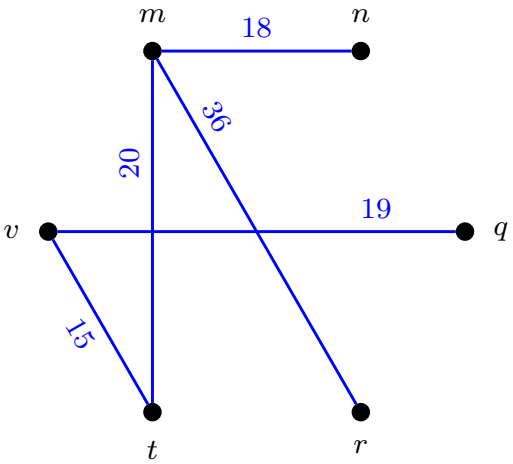
- The next smallest weight is 19, corresponding to edge qv.
- Next is edge mt with weight 20.
- The next smallest weight is 25, corresponding to edge qt, but we must skip it because it would create a cycle.
- The next edge is mq with weight 35, which we also skip.
- The next available edge is mr with weight 36, so we highlight mr.

	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



Kruskal's Algorithm – example

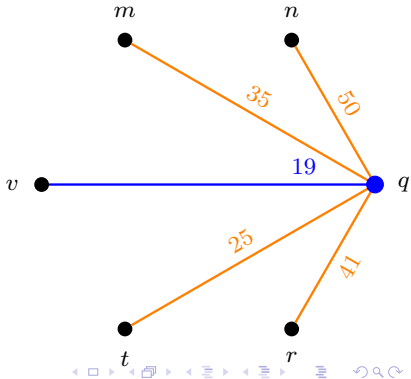
- We have obtained a spanning tree of total weight 108.



Prim's Algorithm – example

- Let us solve the same problem using Prim's Algorithm.
- The drawing of the underlying complete graph is omitted.
- Let us start at vertex q.
- The smallest edge incident to q is qv.

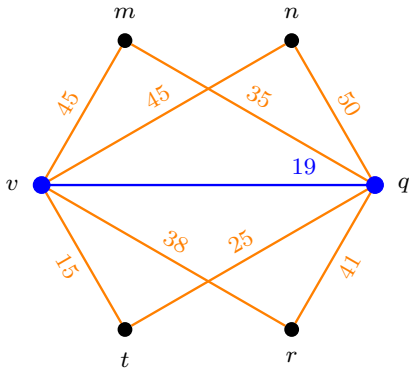
	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



Prim's Algorithm – example

- Next, search for the smallest-weight edge with exactly one endpoint in the current tree, whose vertices are q and v.
- The next edge is vt.

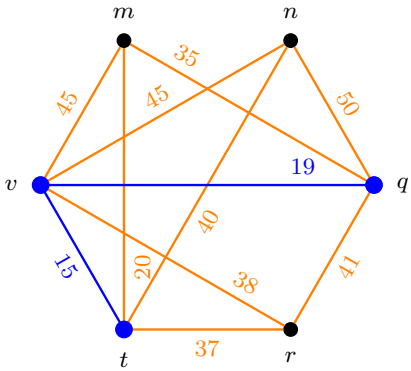
	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



Prim's Algorithm – example

- Next, search for the smallest-weight edge with exactly one endpoint in the current tree, whose vertices are q, t, and v.
- The next edge is tm.

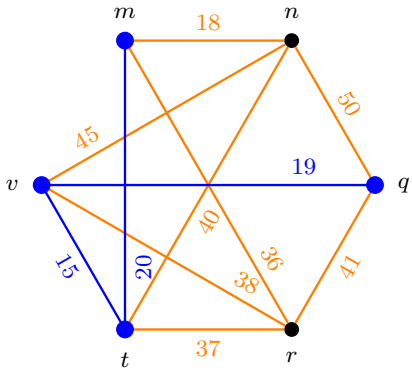
	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



Prim's Algorithm – example

- Next, search for the smallest-weight edge with exactly one endpoint in the current tree, whose vertices are q, t, v, and m.
- The next edge is mn.

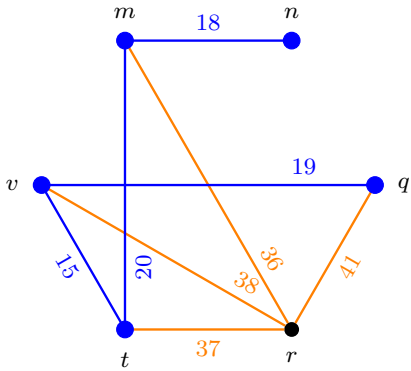
	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



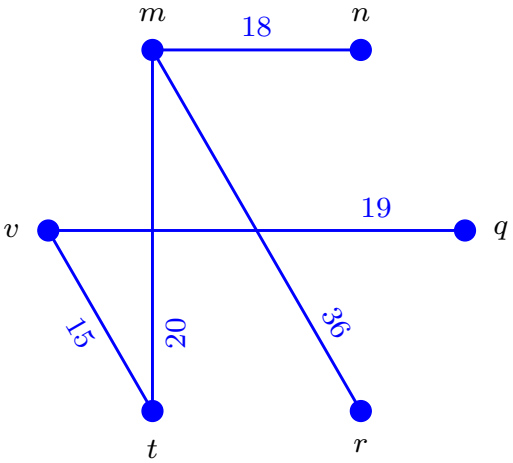
Prim's Algorithm – example

- There is only one vertex left to add to the spanning tree, so we must choose the smallest available edge incident to r.
- The next edge is mr.

	Mesa	Natick	Quechee	Rutland	Tempe	Vinton
Mesa	·	18	35	36	20	45
Natick	18	·	50	42	40	45
Quechee	35	50	·	41	25	19
Rutland	36	42	41	·	37	38
Tempe	20	40	25	37	·	15
Vinton	45	45	19	38	15	·



Prim's Algorithm – example



- The MST has total weight 108.